

PitchVision: Broadcast to 2D Mini-Map Projection

Computer Vision Course Project

Peter Yaacoub

Project Type: Productivity

June 18, 2026

Project Motivation & Goal

Problem Statement

Broadcast cameras provide excellent viewing perspective but fail to explicitly convey the full tactical layout and player configurations across a pitch.

Project Goal:

Build an experimental, unsupervised pipeline that takes standard broadcast frames, processes them, and projects player coordinates onto a standardized 2D tactical mini-map.

- No camera calibration metrics provided beforehand.
- Operates without verified ground-truth annotations.
- Identifies explicit failure modes and practical limitations of unsupervised frameworks.

To keep the pipeline strictly focused on core Computer Vision components:

What PitchVision Does:

- Detects players and referee via YOLO.
- Clusters jerseys by color groups.
- Generates homography matrices from field line segmentations.
- Tracks player ids across consecutive frames.

Explicit Non-Goals:

- No predictive modeling on future movements/passes.
- No multi-camera sync (single continuous shot focus).
- Does not try to create perfect reconstructions over entire matches.

The Implemented Pipeline

The framework operates sequentially on raw video inputs:

- 1 **Frame Extraction:** Sampled evenly spaced subsets (for debugging) or sequential intervals (for final analysis).
- 2 **Object Detection:** Deep learning (YOLO) identifies players, ball, and referees.
- 3 **Jersey-Color Clustering:** Masking out the grass backdrop and identifying team boundaries via K-Means.
- 4 **Field Line Extraction:** Canny Edge + Hough Transform to determine layout indicators.
- 5 **Approximate Projection:** Direct transformation matrix mapping frame boundaries onto a 2D plane.
- 6 **Tracking:** Smooth frame-to-frame correlations with persistent track IDs.

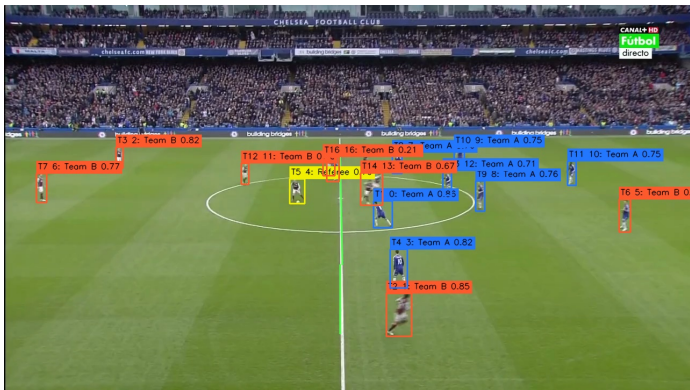
Detection & Team Clustering

- **YoloDetector:** Crops bounding boxes around players and filters them based on expected bounding box area parameters.
- **Grass Masking:** Uses an HSV range filter (35–95 Hue) to isolate and mask out field green, protecting color integrity.
- **Upper Body Isolation:** Crops to the upper 62% of the player box to minimize socks/boots interference.
- **Clustering Strategy:** Employs K-Means to identify primary team configurations while handling exceptions for referees and goalkeepers.

Pitch Line Extraction & Homography

Establishing spatial coordinates without fixed calibrations:

- **Field Segments:** Extracted via morphology operators on green masks.
- **Line Detection:** Hough Transform (HoughLinesP) isolates white pitch markings.
- **Homography Estimation:** Map visual boundaries to standard 105m $\times$ 68m definitions.



The SimpleTracker Framework

KISS Paradigm Implementation

Swapped complex Hungarian/Kalman setups for a light, deterministic, dependency-free tracker relying on **constant-velocity predictions** and **greedy assignment**.

Assignment Cost Metric:

$$\text{Cost} = \text{Spatial Distance} + \text{Team Penalty} + \text{Color Penalty}$$

- **Velocity Vector:** Rolling momentum: $v_{new} = 0.6 \cdot v_{curr} + 0.4 \cdot v_{old}$.
- **Persistence:** Tracks persist up to 6 missing frames to handle temporary occlusion.

Experimental Setup & Datasets

The code supports subcommands: `extract-frames`, `run-checkpoint`, and `run`.

- **50% Milestone Subset:** 50 static frames for validating detection and clustering.
- **Final Target Segments:** Continuous video (10–15s) for evaluating panning artifacts and tracking consistency.

Metrics: Team clustering F1-scores and relative projection stability.

Milestone Progression & Visualization

50% Checkpoint Status:

- Reliable player bounding boxes.
- Upper torso extraction resolved class mixing.
- Ball detection thresholds optimized.

Final Implementation Outputs:

- Stitched side-by-side videos.
- Live updating mini-maps.
- Unified logging via JSON.

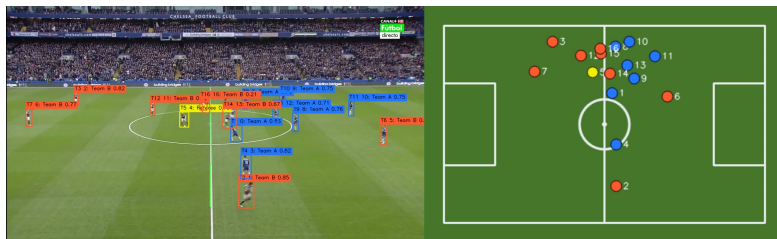


Figure: Side-by-side view: Broadcast overlay vs tactical tracking mini-map.

Observed Failure Modes

- **Projection Jitter:** Slight changes in line detections skew the homography matrix calculation frame-to-frame.
- **Jersey Mixing:** Shadows, movement, or kits close to green create clustering edge cases.